TENSORING A PROJECTIVE MODULE WITH A G-MODULE ALONG THE DIAGONAL ACTION STAYS PROJECTIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Hom-tensor adjunction with diagonal G-action: Hom_{Lambda[G]}(P tensor_A M, N) = Hom_{A[G]}(P, Hom_Lambda(M, N)) with the conjugate action (g.phi)(m) = g phi(g^{-1}m). invoked at proof lines 1704-1711. Reduces exactness of Hom_{Lambda[G]}(P tensor M, -) to the displayed composite.

Lemma 0.2. *Let P be a finite projective $A[G]$-module and M a $\Lambda[G]$-module, finite projective as a Λ-module. Then $P \otimes_A M$ is a finite projective $\Lambda[G]$-module, for the structure induced by the diagonal action of G.*

Proof. For any $\Lambda[G]$-module N one has

$$\mathrm{Hom}_{\Lambda[G]}\left(P \otimes_A M, N\right) = \mathrm{Hom}_{A[G]}\left(P, \mathrm{Hom}_{\Lambda}(M, N)\right)$$

where the G-action on $\mathrm{Hom}_\Lambda(M, N)$ is given by $(g \cdot \varphi)(m) = g\varphi(g^{-1}m)$. Now it suffices to observe that the right-hand side is a composition of exact functors, because of the projectivity of P and M. □

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